Perturbation-informed mapping reveals *RAC1*-responsive tumor-state organization in lung adenocarcinoma

Research Project Brief | Computational Biology / Single-cell Perturbation Analysis

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GitHub: https://github.com/likejin6/RAC1-latent-state-mapping

Summary

This project converts *RAC1* virtual perturbation-derived GRN rewiring information into graph-informed features for single-cell tumor-state mapping,revealing hybrid functional states and continuous program gradients in LUAD.A Python-based graph representation analysis further support network-role shifts of DR genes in networks

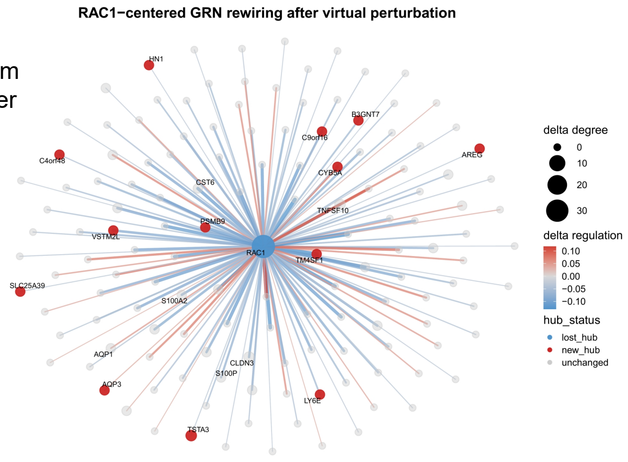

Fig. 1. *RAC1*-centered GRN rewiring after virtual perturbation.

Research question

Can *RAC1*-responsive GRN rewiring be used to interpret latent tumor-state organization in lung adenocarcinoma?

Motivation

LUAD tumor cells exhibit heterogeneous and plastic functional states, but the regulatory network-level organization underlying these latent states remains incompletely understood. Conventional differential expression analysis may miss perturbation-associated regulatory rewiring and gene network-role changes.

Computational framework

1. **Candidate prioritization & virtual knockout:** Network hub scoring prioritizes candidate targets; scTenifoldKnk performs in silico knockout to infer GRN rewiring.
2. ***RAC1*-specific GRN rewiring:** WT and *RAC1*-vKO networks are compared to identify rewired regulatory interactions and hub-status shifts.
3. **Perturbation-informed manifold :** Top *RAC1*-responsive genes selected by network-role changes are used to reconstruct a latent tumor-state manifold via PCA/UMAP.
4. **Functional state mapping:** UCell scored six curated programs and annotated hybrid tumor states and functional gradients.
5. **Clinical relevance:** *RAC1*-responsive genes are projected to TCGA-LUAD for prognostic modeling.
6. **Supplementary graph validation:**WT/RAC1-vKO GRNs are analyzed in Python using topology metrics, adjacency-vector PCA, and node2vec embeddings.

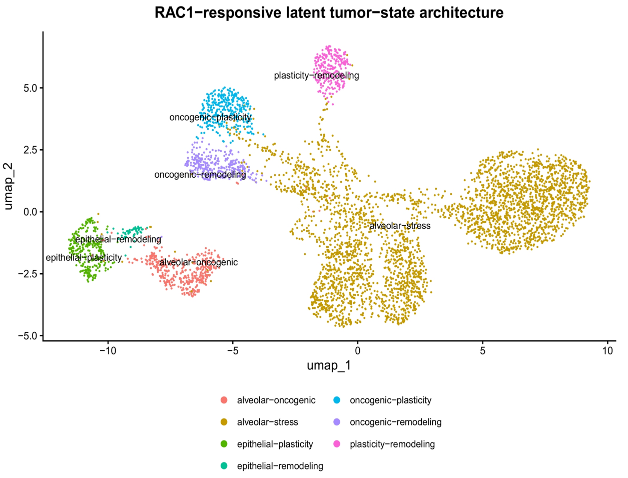

Fig. 2. Perturbation-informed latent tumor-state manifold.

Key findings

1. ***RAC1* showed the strongest virtual perturbation response** among 10 targets, with the highest mean distance, max distance and number of significant DR genes.
2. ***RAC1*-vKO reshaped GRN topology.** producing a 141-gene / 140-edge rewiring network with 13 new hubs and 1 lost hub.
3. **99 *RAC1*-responsive genes reconstructed a perturbation-informed latent manifold** with 10 clusters and 7 hybrid functional states.
4. **Functional program scoring** revealed hybrid tumor-state organization and continuous program gradients, while an 8-gene RAC1-responsive signature retained **prognostic relevance** in TCGA - LUAD.
5. **Python-based graph analysis** showed **greater** topology and embedding **shifts in DR genes** than in non-DR genes, supporting network-role rewiring.

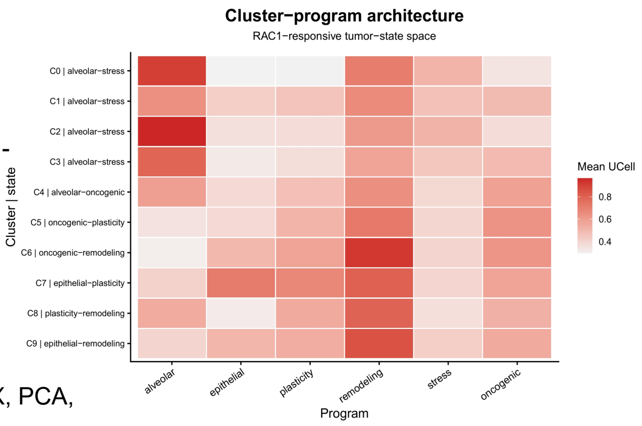

Fig. 3. Cluster-program architecture across *RAC1*-responsive latent states.

Skills demonstrated

- Single-cell transcriptomic analysis and tumor-cell subset extraction
- scTenifoldKnk-based virtual perturbation and GRN rewiring analysis
- Network topology and graph representation analysis using NetworkX, PCA, and node2vec
- Perturbation-informed feature construction for latent state mapping
- Low-dimensional manifold reconstruction using PCA/UMAP
- UCell-based functional program scoring and hybrid state annotation
- TCGA-LUAD survival modeling: LASSO-Cox, Kaplan–Meier, time-dependent ROC, C-index
- Modular R/Python workflow and GitHub-based reproducible documentation

Project positioning

This project is not a conventional biomarker discovery study. It demonstrates a computational perturbation biology strategy linking virtual perturbation-derived regulatory rewiring to systems-level tumor-state mapping in LUAD.

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